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solver for Example 8.10. Each solver reduces the size of the input by one and passes it on to a sub-solver, resulting in 5 solvers. This continues until the input received by a solver is small enough to output the solution directly. The last solver received [E] as the input. Since [E] is small enough, the solver outputs the length of [E] as 1 immediately, and the recursion stops.

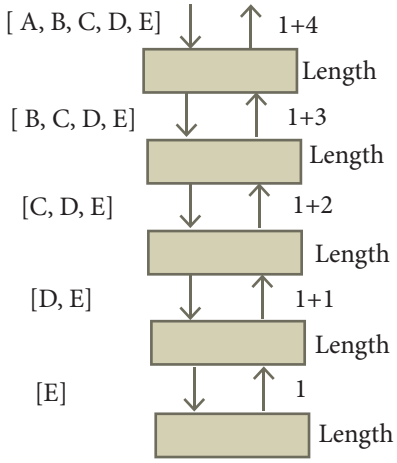

Figure 8.6: Recursive process with solvers and sub-solvers

The recursive process for length [A,B,C,D,E] is shown in Figure 8.7.

```
1     length [A,B,C,D,E]
2   = 1 + 1 length [B,C,D,E]
3   = 1 +1 + length [C,D,E]
4   = 1 + 1+ 1 + length [D,E]
5   = 1 + 1+ 1 + 1 + length [E]
6   = 1 + 1 + 1 + 1 +1
7   = 1 +1 +1 + 2
8   = 1 + 1 + 3
9   = 1 + 4
10  = 5
```

Figure 8.7: Recursive process for computing the length of a sequence

8.4.2 Recursive problem solving

Each solver should test the size of the input. If the size is small enough, the solver should output the solution to the problem directly. If the size is not small enough, the solver should reduce the size of the input and call a sub-solver to solve the problem with the reduced input. For Example 8.10, solver's algorithm can be expressed as

$$\text{length of sequesce} = \begin{cases} 1 & \text{if sequence has only one customer} \\ 1 + \text{length of tail}, & \text{otherwise} \end{cases}$$